Fundermentals of R

Intermediate R - 1.0 Data

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Introduction

R provides a wide range of built-in datasets that are readily available for analysis and exploration. These datasets cover various domains, including statistics, economics, biology, and social sciences. In this lesson, we will explore some commonly used built-in datasets in R along with code examples to access and manipulate them

In Built Data

R comes with a package called "datasets" that contains numerous built-in datasets. To access these datasets, we have to load the "datasets" package. To view a list of all the datasets, you can use either `data()` or `library(help = "datasets")`, run the code chunks below to see them in action.

NOTE: This lesson is only limited to showing how to loading data, manipulation will come later on.

```
#library(help = "datasets")
```

```
#data()
```

Sample npk Data

```
# view the first 4 rows
head(npk, 4)
```

```
##   block N P K yield
## 1     1 0 1 1  49.5
## 2     1 1 1 0  62.8
## 3     1 0 0 0  46.8
## 4     1 1 0 1  57.0
```

Sample Cars Data

```r
# view the first 4 rows
head(cars, 4)
```

```
##   speed dist
## 1     4    2
## 2     4   10
## 3     7    4
## 4     7   22
```

User Created Data

In R, users have the flexibility to create and manipulate their own data sets. This capability allows for the exploration, analysis, and modeling of custom data, tailored to specific needs. In this lesson, we will cover various techniques and code examples for creating user-created data in R.

Vectors

```r
# Creating a numeric vector
num_vec <- c(1, 2, 3, 4, 5)
num_vec
```

```
## [1] 1 2 3 4 5
```

```r
# Creating a character vector
char_vec <- c("one", "two", "three")
char_vec
```

```
## [1] "one"   "two"   "three"
```

```r
# Creating a logical vector
logic_vec <- c(TRUE, FALSE, TRUE)
logic_vec
```

```
## [1]  TRUE FALSE  TRUE
```

Data Frame

```r
# Creating a data frame
fname <- c("James", "Emily", "David")
age <- c(25, 30, 35)

df <- data.frame(fname, age)
df
```

```
##   fname age
## 1 James  25
## 2 Emily  30
## 3 David  35
```

Matrix

```r
# Creating a matrix
mat <- matrix(1:9, nrow = 3, ncol = 3)
mat
```

```
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
```

Lists

```r
# Creating a list
lis <- list(num_vec, char_vec, df, mat)
lis
```

```
## [[1]]
## [1] 1 2 3 4 5
##
## [[2]]
## [1] "one"   "two"   "three"
##
## [[3]]
##   fname age
## 1 James  25
## 2 Emily  30
## 3 David  35
##
## [[4]]
##      [,1] [,2] [,3]
## [1,]    1    4    7
## [2,]    2    5    8
## [3,]    3    6    9
```

Factors

```r
# Creating a factor
gender <- c("Male", "Female", "Male", "Male")
factor_gender <- factor(gender)

factor_gender
```

```
## [1] Male   Female Male   Male
## Levels: Female Male
```

Working Directory

The working directory in R refers to the folder or directory where R will look for files and where it will save files by default. Understanding and managing the working directory is important for efficient file handling and data analysis. In this lesson, we will cover the concept of the working directory in R and explore various functions to set, check, and change the working directory.

Get Working Directory

To check the current working directory in R, you can use the `getwd()` function. This function returns the path to the current working directory as a character string. Run the code chunk to get the current working directory.

```r
# get current working directory
getwd()
```

```
## [1] "F:/Software Data Solutions/Courses/r-course/Intermediate/1.0 Data"
```

Setting the working Directory

To set a specific directory as the working directory in R, you can use the `setwd()` function or use the IDE setting. This function takes a path to the desired directory as an argument and sets it as the working directory.

```r
# set a working directory
#setwd("F:/R/r-course/Intermediate")
#getwd()
```

List Files in working directory

```r
list.files()
```

```
## [1] "1.0-Data-Part-1.html" "1.0-Data-Part-1.pdf"  "1.0-Data-Part-1.Rmd"
## [4] "1.0 Data Part 1.Rmd"  "data"
```

Load External Data

Lets first create own files to use for this section of the lesson. We will create a data folder, and the write our various data into the folder.

Create Data

Run the code chunk below to create various data files

```
# create the data to be written
data = data.frame(name=c("James John", "Sydney Aundrew", "Esther Kubuni"),
                  age = c(23, 34, 67),
                  gender = c("M","M","F"))
# create csv data types
# two write the data in the current folder, we have to change the working directory
#setwd("F:/R/r-course/Intermediate/1.0 Data")

# write csv file
# create the data folder
write.csv(data, file = "./data/data.csv")

# write txt file
write.table(data, file = "./data/data.txt")

# write xlsx file
write.xlsx(data, file = "./data/data.xlsx")

# write json file
jsonlite::write_json(data, path = "./data/data.json")

# test

# for now we will work with these, the rest you'll work with during the assignement
```